HW8: Tree-Based Methods

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1 Objective and Setup

This report studies how tree-based models learn structure from data in both regression and classification settings. The regression analysis uses the Ames Housing dataset, where the response is `SalePrice`. The classification analysis uses the Breast Cancer Wisconsin dataset, where the goal is to distinguish malignant and benign tumors. The main themes are how trees partition the feature space, how stable those partitions remain under perturbation, how ensemble methods improve predictive performance, and how BART differs from standard tree ensembles.

For the Ames data, I removed the identifier columns `Id` and `PID`, median-imputed numeric variables, mode-imputed categorical variables, and one-hot encoded the categorical predictors. A common train-test split with `random_state = 42` was used throughout the regression tasks so that all model comparisons refer to the same held-out observations. For the breast cancer data, I used a stratified train-test split with the same random state.

2 Task 1: Tree Structure, Splits, and Interactions in Regression

A fully grown decision tree regressor was fit on the Ames training split. The root split was

$$\texttt{Overall Qual} \leq 7.5,$$

which already captures the main structure of the dataset: housing quality is the strongest first-order separator of price. Across the top three levels in Figure 1, the only features that appear are `Overall Qual` and `Gr Liv Area`, so the early part of the tree first creates broad quality bands and then refines them using effective living space.

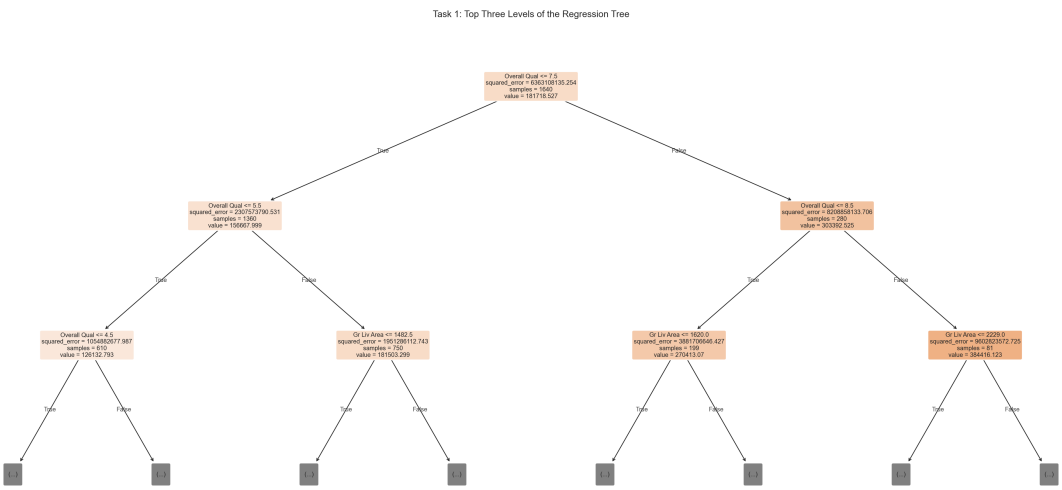

Figure 1: Top three levels of the Ames regression tree.

To interpret root-to-leaf paths in a more meaningful way, I also used a shallow companion tree with the same dominant top split. Two representative subgroups emerged clearly:

- **High-end houses:** `Overall Qual > 8.5` and `Gr Liv Area > 2229`, with 36 observations and mean sale price about 439713.61.
- **Lower-priced houses:** `Overall Qual <= 4.5` and `Total Bsmt SF <= 799.5`, with 90 observations and mean sale price about 86017.77.

Written as explicit root-to-leaf rules, the two paths are:

Decision rules	Subset description	n	Mean	Median
`Overall Qual > 7.5` → `Overall Qual > 8.5` → `Gr Liv Area > 2229`	large, high-quality homes in the training split	36	439713.61	458673.50
`Overall Qual <= 7.5` → `Overall Qual <= 5.5` → `Overall Qual <= 4.5` → `Total Bsmt SF <= 799.5`	lower-quality homes with smaller basements in the training split	90	86017.77	85750.00

Table 1: Two representative root-to-leaf paths from the interpretable companion tree.

These paths show the interaction structure naturally captured by trees. Living area matters differently once quality is already high, while basement size is more relevant among lower-quality homes. A linear model would need manually specified interaction terms to express the same conditional logic. On this split, however, linear regression still generalized better numerically: the tree had test RMSE about 33246.80, while linear regression had test RMSE about 20984.32. So the tree is more interpretable locally, but the linear model was more accurate here.

3 Task 2: Instability and Model Complexity

To study instability, I compared three regression trees on the same test set: a deep unconstrained tree, a shallow pruned tree, and a deep tree fit on a slightly perturbed training sample. Their overall complexity is summarized in Table 2.

Model	Depth	Leaves	Test RMSE
Deep tree	26	1560	33246.80
Shallow tree	4	15	35722.61
Perturbed deep tree	24	1421	33527.31

Table 2: Complexity and test error for the three trees used in the instability analysis.

The most important split remained stable: all three trees began with `Overall Qual`, and the first two levels followed the same pattern with thresholds at 7.5 at the root, 5.5 on the lower-quality branch, and 8.5 on the higher-quality branch. But below those early splits, the structures diverged noticeably. The deep tree fit the training data perfectly, while the perturbed deep tree produced a different lower-level partition with almost the same test error. The shallow tree was much easier to read, but its predictive accuracy dropped.

The instability is also visible in the predictions themselves. The mean pointwise standard deviation across the three test-set prediction vectors was about 18458.99, which is large in housing-price units. This is the essential high-variance behavior of decision trees: small data changes do not necessarily alter the main story at the top of the tree, but they can still rewrite many lower-level rules and move predictions substantially.

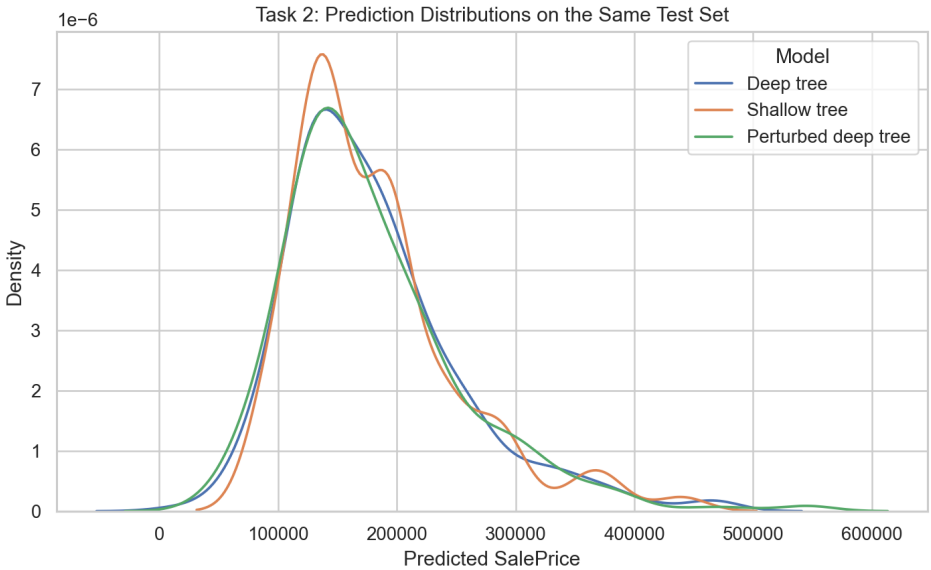

Figure 2: Prediction distributions from the deep, shallow, and perturbed trees on the same test set.

Figure 2 shows that this structural instability carries through to the fitted values as well. Tree depth therefore affects all three practical aspects of the fit. Greater depth improves flexibility and can reduce bias, but it also makes the fitted partition more fragile. Pruning improves interpretability and stability, though usually at some predictive cost.

4 Task 3: Classification

A decision tree classifier with depth 5 and minimum leaf size 5 was fit to the Breast Cancer Wisconsin data. Its test accuracy was 0.9211, so the model performed well overall. The more interesting part is how the tree expresses confidence.

Three representative test observations illustrate that behavior. Table 3 records the leaf assignment, the empirical class proportions at that leaf, and the leaf sample size used to form the probability.

Case	Test index	Leaf node	Leaf sample size	Class proportions
High-confidence correct	256	22	141	malignant = 1.0, benign = 0.0
Low-confidence correct	363	21	5	malignant = 0.4, benign = 0.6
High-confidence incorrect	406	22	141	malignant = 1.0, benign = 0.0

Table 3: Representative classification cases and their terminal leaves.

This follows directly from how classification trees form probabilities. The predicted probability at a leaf is simply the empirical class proportion in that leaf. A probability of 0.6 means the leaf contains 60% training observations from that class. A probability of 1.0 means the leaf

is pure in the training data, even if a test point landing there happens to belong to the other class.

That also explains why trees can be badly calibrated. Their probabilities are piecewise constant and can change abruptly from one region to the next. Logistic regression provides a useful contrast because it models probabilities directly rather than assigning one empirical proportion to an entire leaf. The Brier scores reflected this difference:

- Decision tree: 0.0692
- Logistic regression: 0.0284

Misclassifications are therefore not only about wrong labels; they also reveal how the geometry of the tree affects uncertainty. The highlighted mistake is especially instructive because it does not occur in a tiny mixed leaf. It falls in leaf 22, which is large and pure in the training data, so the error comes from an overconfident regional rule rather than from a noisy, low-sample terminal node. Small mixed leaves still produce low-confidence predictions, but they are not the only source of mistakes.

5 Task 4: Ensemble Methods: Random Forest vs Gradient Boosting

The single tree from the earlier regression analysis was compared with a Random Forest and a Gradient Boosting Regressor on the same Ames split. The training and test RMSE values are shown in Table 4.

Model	Train RMSE	Test RMSE
Decision tree	0.00	33246.80
Random forest	12186.08	23128.03
Gradient boosting	12821.22	20102.34

Table 4: Training and test errors for the single tree and the two ensembles.

Both ensembles improved strongly over the single tree. The decision tree overfit completely, while Random Forest and Gradient Boosting kept low training error and generalized much better. Among the three, Gradient Boosting performed best on the held-out set.

The feature-importance rankings from the two ensembles were also broadly consistent. Both emphasized `Overall Qual` most strongly, followed by size-related variables such as `Gr Liv Area`, `Total Bsmt SF`, `1st Flr SF`, and garage-related features, as shown in Figure 3. That agreement suggests that the underlying predictive structure is not arbitrary even though the single-tree partitions are unstable.

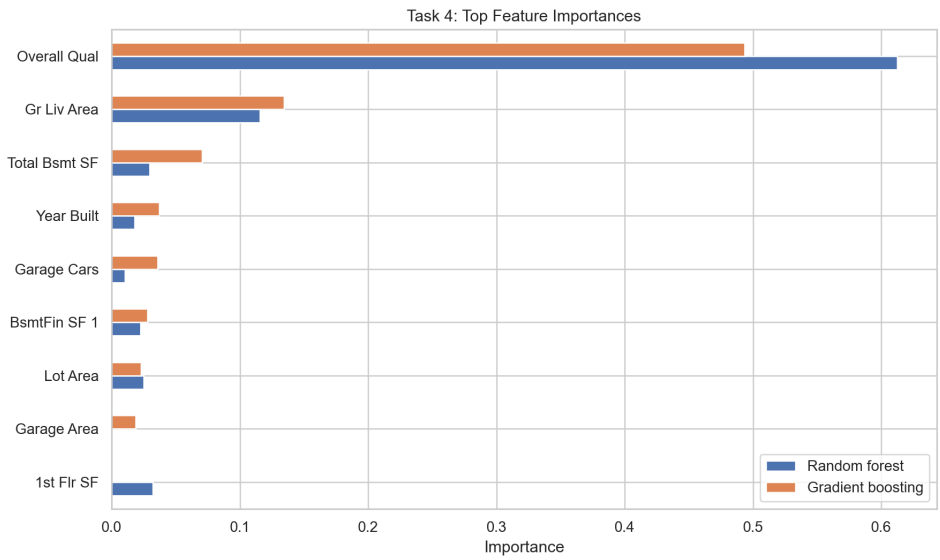

Figure 3: Top feature importances from Random Forest and Gradient Boosting.

The benefit of the ensembles becomes especially visible on the observations where the single tree performs poorly. Table 5 lists the five worst test cases for the single tree. In each case, at least one ensemble moves substantially closer to the true sale price. The single tree makes a hard regional decision; the ensembles borrow strength from many trees and therefore behave less erratically on these difficult cases.

Actual	Decision tree	Random forest	Gradient boosting	Tree abs. error
415000	209000	244508	255474	206000
257076	122000	237649	225315	135076
345000	475000	321587	322667	130000
151400	279900	219874	149752	128500
383970	260000	271550	304319	123970

Table 5: Subset of test observations on which the single decision tree performs especially poorly.

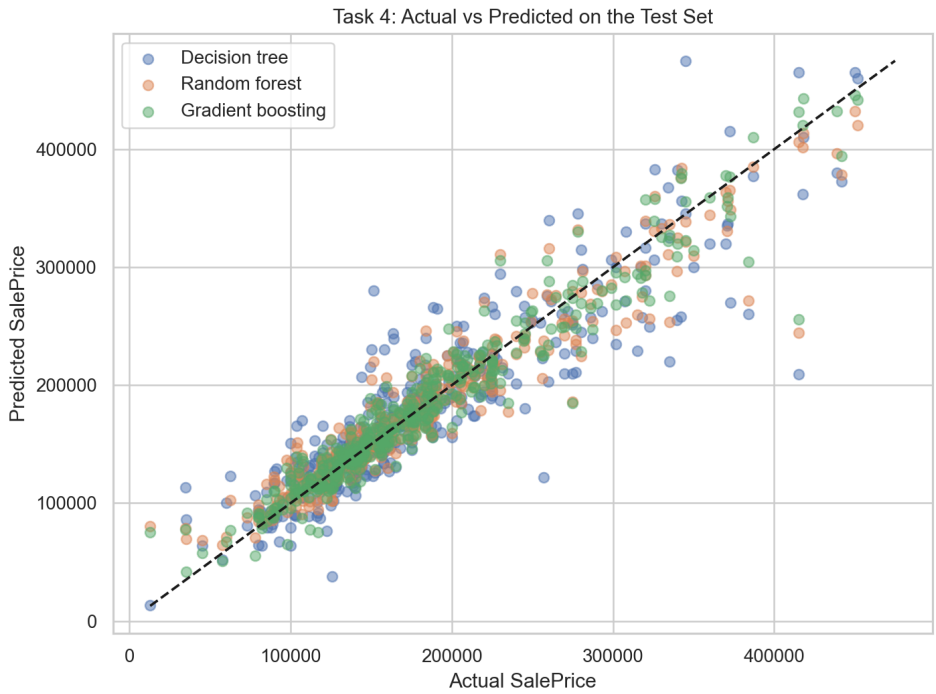

Figure 4: Actual versus predicted sale price for the three regression models on the test set.

Figure 4 shows the same pattern in aggregate: the ensemble predictions stay much closer to the diagonal than the single-tree predictions. The difference between the two ensemble methods is conceptual as well as empirical. Random Forest builds many decorrelated trees in parallel and averages them, so its main role is variance reduction. Gradient Boosting builds trees sequentially to correct residual errors, so it can reduce bias more aggressively. Gradient Boosting is therefore especially attractive when the response has strong structured residual patterns that can be peeled away stage by stage, which fits this housing problem well. Random Forest is often preferable when the signal is rougher, the feature set is noisier, or robustness with less tuning matters more than squeezing out the last reduction in bias.

6 Task 5: Bayesian Additive Regression Trees

6.1 Conceptual Summary

BART models the response as a sum of many weak regression trees under a Bayesian prior. Instead of producing a single fitted ensemble in the usual frequentist sense, it works with a posterior distribution over possible ensembles. The result is not only a predictive mean but also a distribution of plausible predictions, which gives a direct measure of uncertainty.

6.2 Implementation and Comparison

Using the same train-test split as in Task 4, I fit BART on the top 25 Random Forest features from the training split. This restriction kept posterior sampling manageable in the one-hot encoded feature space while preserving the strongest predictive variables. The fitted BART model achieved:

- Test RMSE: 27012.58
- Test R^2: 0.8772

- Mean posterior predictive standard deviation: 17865.28

So BART improved over the single decision tree but did not match the Random Forest or Gradient Boosting models on this split. In pure predictive terms, it sits between the single tree and the stronger frequentist ensembles.

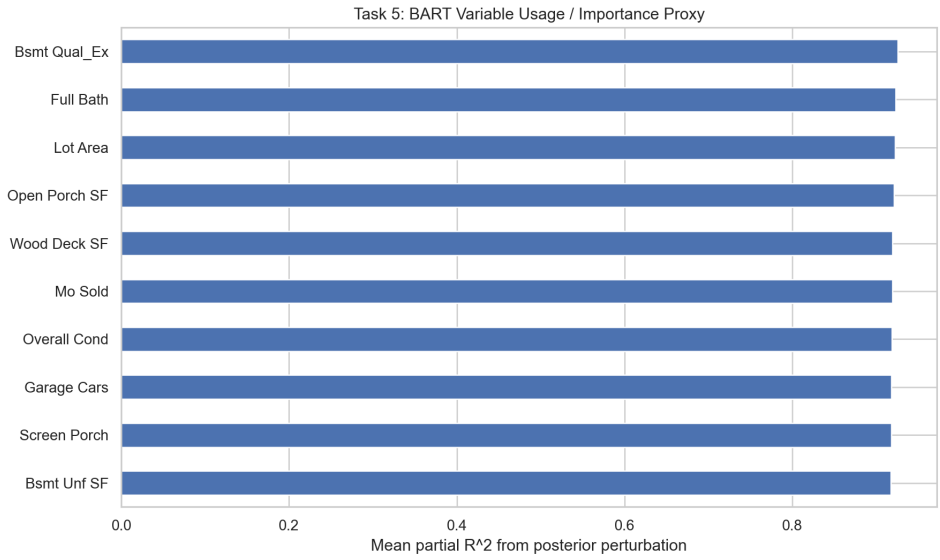

Figure 5: BART variable-usage / importance proxy on the top-25 feature subset.

Figure 5 shows that the BART usage proxy is spread across a wider set of predictors than the impurity-based ensemble importances. Variables such as `Mas Vnr Area`, `Open Porch SF`, `Kitchen Qual_Ex`, `Exter Qual_Ex`, `Garage Area`, and `Year Remod/Add` all receive substantial usage. Compared with Random Forest and Gradient Boosting, where `Overall Qual` and `Gr Liv Area` dominate more strongly, the BART fit distributes signal across correlated finish, amenity, and renovation variables. That is consistent with an additive collection of weak trees sharing explanatory burden rather than forcing a small number of variables to drive the entire split hierarchy. The main gain from BART is not only its point prediction but the uncertainty attached to that prediction. The posterior predictive standard deviation summarizes how variable the prediction remains after seeing the training data. That matters in applications such as pricing, lending, medical prognosis, or policy decisions, where knowing how uncertain a prediction is can be as important as the prediction itself.

7 Conclusion

The overall picture is clear. A single decision tree provides the most transparent local rules, but its fitted structure is unstable and its predictive performance is weaker than the ensemble alternatives. Random Forest improves substantially by averaging many trees and reducing variance. Gradient Boosting performs best on this dataset because the sequential residual corrections capture structure that a single tree misses. BART offers a different tradeoff: it is competitive, though not best here, and its distinguishing advantage is posterior predictive uncertainty.

Taken together, the analysis shows that tree-based methods should be judged on more than one axis. Interpretability, stability, predictive accuracy, and uncertainty quantification do not all favor the same model. On this dataset, Gradient Boosting is the strongest predictive choice, while BART is the most informative when uncertainty matters and a single tree remains the easiest model to explain.